

IS 3920 – Individual Project on Business Solutions

Software Requirements Specification

for

Gem Collector's Portfolio and Marketplace

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Student's Declaration

I hereby declare that this report is my original work and has not been submitted, in whole or in part, for any degree or diploma at any university or other institution of higher learning. All information derived from the work of others, whether published or unpublished, has been properly acknowledged in the text, and a complete list of references has been provided.

.....2026-01-18.....
Date


.....
Signature of Student

Supervisors' declaration

I hereby declare that I have reviewed this project and find it adequate in both scope and quality.

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Revision History

Name	Date	Reason for Changes	Version

Chapter 1 Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to outline the comprehensive and specific functional and non-functional requirements of the Gem Collector's Portfolio and Marketplace System. Specifically, this system will enable the creation of a secured and transparent online environment wherein gem sellers, collectors, and buyers are able to safely and openly interact. Additionally, this system will guarantee that all gems uploaded within the site are indeed legitimate and certified, and that all occurrences are closely supervised and verified by system administrators prior to being posted online.

The Portfolio and Marketplace System for the Gem Collector is a response to the need for trust, and the absence thereof, and a proper verification process within the gem trading market among online traders. This SRS document goes a long way in making the requirements clear and providing a guideline for the designers, developers, and test engineers to follow, so a common goal is achieved in terms of functionality, constraints, and behavior. This document is the basis for quality and testing, and possibly enhancement, to a certain degree, and helps in eliminating ambiguities in the process of developing the system.

1.2 Document Conventions

This SRS document is prepared according to the IEEE Std 830-1998 standard for Software Requirements Specifications, which provide guidelines, recognized by the industry, that are to be used in writing precise and structured requirements documentation. Formatting conventions used in this document are as follows:

- Section titles and headings are in bold, font size 18, so that major sections are well distinguished.
- Subheadings are font size 14, thus making every subdivision of a chapter stand out and clearly visible.
- Font size 12 has been used for paragraphs and body text, large enough to be easily readable while still looking professional.
- Lists, bullet points and numbered items are consistently formatted to enhance clarity and on-page organization.

These are conventions that have been put in place to enhance readability, but also for ease of reference on the part of the technical reader, the project evaluator, and the future maintainer.

Following a standard structure ensures that every section in the document has a well-stated purpose, and no critical information will be missed.

1.3 Intended Audience and Reading Suggestions

This SRS document is intended for a diverse audience that includes both technical and non-technical stakeholders. The primary readers include:

- **Project Supervisors and Evaluators:** To assess the feasibility, scope, and clarity of the system requirements, ensuring that the proposed system aligns with industry standards and meets academic or organizational objectives.
- **Developers:** To gain a detailed understanding of the system's functional and non-functional requirements, architecture, and user role specifications. Developers will use this information to design, implement, and integrate system modules effectively.
- **Testers:** To design comprehensive test cases, verify system behavior, and ensure that both functional and non-functional requirements are fully satisfied.
- **Future Maintainers:** To understand the system's behavior, constraints, and workflows, allowing for smooth updates, troubleshooting, and expansion of system functionalities in the future.

For readers seeking a general overview of the system, it is recommended to start with Section 1 (Introduction) and Section 2 (Overall Description). Technical readers, including developers and testers, should focus on System Features, External Interface Requirements, and Non-Functional Requirements, where detailed descriptions of system behavior, interfaces, and performance constraints are provided.

1.4 Product Scope

The Gem Collector's Portfolio and Marketplace are a web-based software system developed for authenticated and transparent gem trading in the virtual environment. The system shall focus on providing a professional platform where the seller or collector can upload, manage, and showcase certified gems, while buyers can access verified listings and participate in auctions with confidence.

Three main user roles are the basis of the system: Sellers/Collectors, Buyers, and Administrators. The given system allows sellers and collectors to publish detailed information about their gems: the type of gem, carat, cut, clarity, origin, photographs, and official certificates by the Gem & Jewelry Authority. After being reviewed and approved by administrators, such publications can be viewed by buyers, ensuring that the information displayed is reliable. It is the job of the

administrators to manage user portfolios, check gem authenticities, check the details in the certificates, and allow the listings to be posted for public view.

“The scope of the project shall not include” financial transactions, “automated certificate verification,” and “logistic management,” and instead include authentication, “authorization, portfolio management, data handling and verification, and auction management.” By thus delimiting the scope of the project, the system is ensured to avoid complexity and yet still offer basic functionalities which address trust and transparency issues in online gem trading.

1.4.1 Problem in brief

There are major issues concerning authenticity, trust, and transparency within the online gem and jewelry industry. Buyers often find difficulty in verifying authenticity of gems and certificates when purchasing online, increasing the chance of fraud. Small-scale sellers and private collectors also face hurdles in reaching a wide audience due to a lack of dedicated and professional digital platforms.

General online solutions exist, but they mostly exhibit partial functionalities like basic catalog listing or auction capabilities; they do not offer a comprehensive certificate validation, portfolio management, and administrator-controlled verification integrated all in one platform. Such shortcomings have kept buyers less confident, reduced the accessibility of the portal to genuine sellers, and increased the fraudulent listings ratio. The proposed system will help address these challenges by integrating all these key features into one integrated platform, hence making the online gem trading process more efficient, trustworthy, and transparent.

1.4.2 Aim and Objectives

Aim:

The aim of this project is to develop a Gem Collector’s Portfolio and Marketplace that ensures authentic, transparent, and accessible online gem trading, leveraging the MERN stack (MongoDB, Express.js, React.js, Node.js) and manual certificate verification processes.

Objectives:

- Enable gem sellers and collectors to upload detailed gem information, including high-quality images and official Gem & Jewelry Authority certificates, to ensure authenticity and traceability.
- Develop a centralized digital portfolio system that allows collectors to manage, organize, and showcase certified gems in a professional and structured manner.
- Provide buyers with access to verified gem listings, complete with detailed seller information and gem specifications, enabling informed purchasing decisions.

- Support auction-based gem listings, allowing sellers to reach a wider audience and buyers to engage competitively, while intentionally excluding online payment integration at this stage.
- Ensure administrative verification and approval of all gem listings prior to public display, guaranteeing that only authentic, reliable, and certified gems are accessible to buyers.

1.5 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications, Institute of Electrical and Electronics Engineers.
- IEEE Std 29148-2018, Systems and Software Engineering — Life Cycle Processes — Requirements Engineering.
- MERN Stack Documentation (MongoDB, Express.js, React.js, Node.js) – providing guidelines for backend, frontend, and database development.

Chapter 2 Overall description

2.1 Product Perspective

The Gem Collector's Portfolio and Marketplace System is a stand-alone, web-based application intended to function individually while taking its place within the larger digital gem and jewelry trading ecosystem. It acts as the central entity among buyers, sellers, and collectors, creating an interface that currently divides online gem trading due to fragmented platforms, unverified listings, and limited access to authenticated certification. The verified and controlled environment of this platform invites trust, transparency, and credibility-ingredients necessary for online transactions involving gemstones.

Functionally, the system introduces an admin-controlled workflow that ensures all gem listings and associated certificates are verified before they go live to the buyers. This reduces not only fraudulent listings but also increases buyer confidence in the authenticity of the gems and certificates. Collectors and sellers can manage their personal gem portfolios, track gem history, and create auction listings while buyers can view verified listings, evaluate gem details, and take part in competitive bidding processes.

Though the scope of the existing system does not include linking with financial/logistics systems, it has been developed in a manner that it is extensible. Future releases may include facilities for automated verification of certificates, secure online transaction processing, and linking with delivery and logistics services so that there is a complete solution from start to finish in gem trading on the online platform. The architecture of the system has modularity so that new features can be added.

2.2 Product Functions

The Gem Collector's Portfolio and Marketplace System provide a range of major functions and capabilities to support the needs of all user roles. The primary features include:

- Secure user registration and authentication with role-based access control: Users must register using valid credentials, and the system assigns roles (Seller/Collector, Buyer, or Administrator) that determine the level of access and permissions.
- Uploading and managing gem details, images, and certification documents: Sellers and collectors can input comprehensive gem information, including type, carat, cut, clarity, origin, and high-quality photographs. They can also attach official certificates issued by recognized authorities to support authenticity.
- Administrative verification and approval of gem listings: Administrators review all uploaded information and certificates to validate authenticity before the listing becomes publicly visible. Listings may be approved, rejected, or sent back for correction.

- Creation and maintenance of personal gem portfolios: Collectors can organize and manage their certified gems in digital portfolios, enabling easy access, tracking of auction history, and presentation for buyers.
- Auction-based gem listings with real-time bidding functionality: Sellers can list gems for auction, and buyers can place bids in real time, creating a competitive and engaging marketplace environment.
- Browsing, searching, and filtering verified gem listings: Buyers can search for gems based on various criteria such as type, certificate authority, price range, and seller ratings, ensuring efficient discovery of desired items.
- Viewing seller profiles and certificate information: Buyers have access to detailed profiles of sellers and collectors, including past activity and feedback, as well as the details of gem certificates, providing transparency and informed decision-making.
- Administrative monitoring, moderation, and system management: Administrators oversee all platform activities, including audit trails, auction moderation, and system configuration, to ensure integrity, compliance, and smooth operation.

Each function is carefully designed to enhance usability, reliability, and security, addressing the unique challenges of online gem trading while providing engaging user experience.

2.3 User Classes and Characteristics

2.3.1 Seller / Collector

Sellers and collectors are the key sources of content in the system. Sellers and collectors are responsible for entering certified gemstone information into the system in order to increase credibility. Sellers and collectors are also responsible for managing their portfolios, posting listings for auctions, and making sure that all facts regarding the gemstone are valid.

The sellers and collectors will interact with the administrators in the verification procedure in order to respond to feedback or requests for supplementary documentation. The sellers and collectors also interact with buyers in the context of the auction procedure in order to respond to questions asked in detail. The sellers and collectors will benefit from a professional setting in order to display their gems since the platform will create a professional setting.

2.3.2 Buyer

The buyers utilize the system primarily to view, assess, and bid on verified gem listings. They can examine gem details, certification information, and seller profiles to make well-informed decisions. Buyers interact indirectly with sellers through auctions but do not upload content or perform online payments within the current scope of the system.

The system helps buyers in their journey of search, filtering, and discovery of gems by providing the search features to locate them on specified criteria such as type, certificate authority, or

auction status. This makes it very efficient and user-friendly for buyers to spot their item of interest and bid on it.

2.3.3 Administrator

The administrators are charged with maintaining the integrity of the platforms as well as adherence to the rules that govern the system. They analyze and authenticate all gem submissions, certificates, as well as auction listings, prior to making them available to the public. The administrators have a broader perspective concerning the entire system, which involves user management, auction activity management, among other parameters.

It plays an essential part in ensuring that there is no fraud involved. It also ensures a good relationship between buyers and sellers. It also facilitates smooth running of the market. Administrators have other duties, which include handling audit trails and reports.

2.4 Operating Environment

The system operates within a web-based environment and is compatible with a range of hardware and software configurations to ensure accessibility and usability. The operating environment includes:

- Client devices: Desktop or laptop computers, supporting both Windows and macOS operating systems.
- Supported web browsers: Google Chrome, Microsoft Edge, and Mozilla Firefox, ensuring broad compatibility and performance.
- Backend server: Node.js runtime environment for processing server-side logic and managing requests.
- Database: MongoDB for storing user data, gem details, auction history, and certificate information.
- Network requirements: A reliable internet connection is necessary for accessing the platform, uploading content, and participating in auctions.
- External tools: Integration with cloud storage services is used for managing gem images and certificate files securely, ensuring efficient retrieval and backup.

This environment ensures that the system is robust, scalable, and capable of supporting multiple concurrent users without performance degradation.

2.5 Design and Implementation Constraints

The system must operate under several constraints, which influence design and implementation:

- Exclusion of online payment processing: Financial transactions are outside the current scope, simplifying system design while focusing on auctions and portfolio management.
- Manual certificate verification: Administrators manually validate all certificates, ensuring accuracy and reliability in the absence of automated verification systems.
- Academic project constraints: Development is limited by timeframes, available resources, and scope as part of an academic project.
- Dependence on third-party cloud storage: File uploads for gem images and certificates rely on external cloud storage services, requiring stable connectivity and proper configuration.
- Compliance with data protection and security practices: The system must adhere to basic security and privacy standards, protecting user information and preventing unauthorized access.

These constraints guide development decisions, ensuring the system is functional, secure, and feasible within current limitations.

2.6 User Documentation

To support effective system use, comprehensive user documentation and support materials will be provided:

- User guides for sellers, buyers, and administrators, describing roles, responsibilities, and system navigation.
- Step-by-step instructions for uploading gem details, attaching certificates, creating auctions, and managing portfolios.
- On-screen validation messages and tooltips to assist users during input and workflow processes, reducing errors and enhancing usability.
- Administrator manuals detailing verification, moderation, and monitoring procedures, ensuring consistent enforcement of system rules and high platform integrity.

The documentation aims to provide clear, structured, and accessible guidance, enabling users of all technical skill levels to utilize the platform efficiently and confidently.

2.7 Assumptions and Dependencies

The system operates under several assumptions and dependencies that affect its design and performance:

- Users are assumed to have reliable internet access and basic familiarity with web applications to interact effectively with the platform.
- Sellers are assumed to upload genuine certificates issued by recognized authorities, which form the basis of system trust and verification.

- The system depends on the availability of cloud storage services, secure hosting infrastructure, and stable network connectivity to ensure that files, images, and auction data are reliably stored and accessible.
- Administrators are assumed to have the necessary expertise to validate certificates and maintain system integrity, as this process is manual in the current scope.

These assumptions and dependencies define the operational context of the system, clarifying the conditions under which the platform is expected to function effectively.

Chapter 3 External Interface Requirements

3.1 User Interfaces

The Gem Collector's Portfolio and Marketplace System have a user-friendly web interface UI, which is accessible with all standard web browsers. The UI is also responsive, intuitive, and role-specific, such that all types of users, viz., Seller/Collector, Buyer, and Administrator, would find it convenient to navigate through and operate. The UI is standardized with regard to its layout, color schemes, fonts, and navigation systems, thereby minimizing confusion among users.

3.1.1. Seller/Collector Interface:

Sellers and collectors are provided with a dedicated dashboard where they can perform all actions related to gem uploading, portfolio management, and auction creation. The dashboard includes:

- Forms for gem details: Users can input comprehensive information about each gem, including type, carat, cut, clarity, origin, and high-quality photographs.
- Certificate attachment functionality: Users can upload official certificates issued by recognized authorities, which are later verified by administrators.
- Portfolio management tools: Sellers can organize gems into categories, track auction history, and manage active listings.
- Auction creation interface: Sellers can create auction listings, define starting bids, set auction duration, and manage ongoing auctions.
- Input validation: The system ensures that all required information is correctly provided before submission, minimizing errors and incomplete listings.


GemFolio


Nima Perera
Collector

Dashboard

My Portfolio

Auctions

Add New Gem

Settings >

Sign Out

Welcome back, Nima

Here's a summary of your gem portfolio and recent activity.

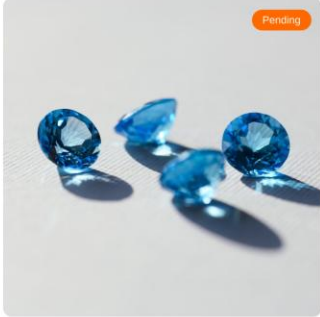
Total Portfolio Value
Rs.63,500
▲ +2.5%

Active Listings
3
out of 4 gems

Pending Verification
0
Awaiting admin review

My Listings

Pending Approved Rejected



Ceylon Blue Sapphire
Carat: 5.2 Origin: Sri Lanka
Type: Sapphire

View Details Edit



Burmese Pigeon Blood Ruby
Carat: 3.8 Origin: Myanmar
Type: Ruby

View Details Edit

Recent Activity

- ✓ Your "Royal Blue Tanzanite" has been approved.
2 hours ago
- 💰 New bid of \$12,500 on your "Sunset Padparadscha".
1 day ago
- ✗ Listing for "Rough Diamond Cluster" was rejected.
3 days ago

Figure 1: Seller/Collector Dashboard

GemFolio

Nima Perera
Collector

Dashboard

My Portfolio

Auctions

+ Add New Gem

Settings >

Sign Out

My Portfolio

Manage your curated collection of certified gems

+ Add New Gem

Search by name, ID..

Filter: All

Sort: Date Added

Gem Image	Name	Weight	GIA #	Status	Visibility	Actions
	Colombian Green Emerald	10.5 ct	GIA #65475	Pending	Visible	edit delete
	Colombian Green Emerald	10.5 ct	GIA #65475	Public	Visible	edit delete
	Colombian Green Emerald	10.5 ct	GIA #65475	Private	Visible	edit delete
	Colombian Green Emerald	10.5 ct	GIA #65475	Public	Visible	edit delete
	Colombian Green Emerald	10.5 ct	GIA #65475	Pending	Visible	edit delete
	Colombian Green Emerald	10.5 ct	GIA #65475	Public	Visible	edit delete
	Colombian Green Emerald	10.5 ct	GIA #65475	Private	Visible	edit delete
	Colombian Green Emerald	10.5 ct	GIA #65475	Public	Visible	edit delete

Figure 2:Collector Portfolio page


GemFolio

Create Your Gem Auction

Set the price and duration for your auction. Your listing will go live once confirmed.

Auction Pricing

Define the financial parameters for your auction

Set your starting price

Rs. 50,000.00

Set minimum bid increment

Rs. 50,000.00

Auction Timing

Choose how long your auction will run

Auction Duration

5 Days

Auction Summary

Review your seeings before going live

Auction Ends on:	October 29, 2025 at 4.30 PM
Listing Fee:	Rs.2500.00
Success Fee:	5% of trial sale price

By starting this auction you agree to the Gemstone Auctions Terms of Services and Auction Policies

Cancel

Confirm and Start Auction



Listing Id: 45BB2-9

Flawless 2.5ct Burmese Ruby

Admin Verified

Carat:2.5, Type: Ruby, Origin: Burma

Figure 3:Create Auction page

3.1.2. Buyer Interface:

The buyer interface is designed to provide easy access to verified gem listings, enabling buyers to evaluate and participate in auctions with confidence. Features include:

- Browsing and searching capabilities: Buyers can search for gems based on type, certificate authority, price range, seller ratings, and other attributes.
- Detailed gem information: Each listing displays comprehensive gem details, high-resolution images, and attached certificates for authenticity verification.
- Auction participation tools: Buyers can place bids, track auction status, and view real-time updates during auctions.
- Filtering and sorting functionality: Buyers can filter listings by criteria such as gem type, certificate status, and auction end time for more efficient discovery.

The screenshot displays the GemFolio Buyer Dashboard for Hasi Perera. The interface includes a top navigation bar with the GemFolio logo, a search bar, and a user profile section. A left sidebar contains navigation links for Dashboard, Marketplace, Auctions, My Watchlist, Settings, and Sign Out. The main content area features a welcome message, three quick-action cards (View Bids, My Watchlist, Recent Dealings), a section for 'Your Watched Auctions' listing two items (Ceylon Blue Sapphire and Colombian Emerald) with their current bids and time left, and a 'Pick Up Where You Left Off' section with three images of gemstones.

GemFolio

Search for gems, auctions...

Hasi Perera
Buyer

Welcome back, Hasi!
You have 3 active bids

Dashboard

- Marketplace
- Auctions
- My Watchlist

View Bids

My Watchlist

Recent Dealings

Your Watched Auctions [View All >](#)

	Current Bid	Bidders	Time Left
Ceylon Blue Sapphire 5.2 Carat	Rs.18,500	14	5d 7h
Colombian Emerald 4.1 Carat	Rs.14,000	24	3d 9h

Pick Up Where You Left Off [View All >](#)

Three images showing gemstones: three blue sapphires, a cluster of red and pink gemstones, and a green emerald being examined with a loupe.

Settings >

Sign Out

Figure 4:Buyer Dashboard

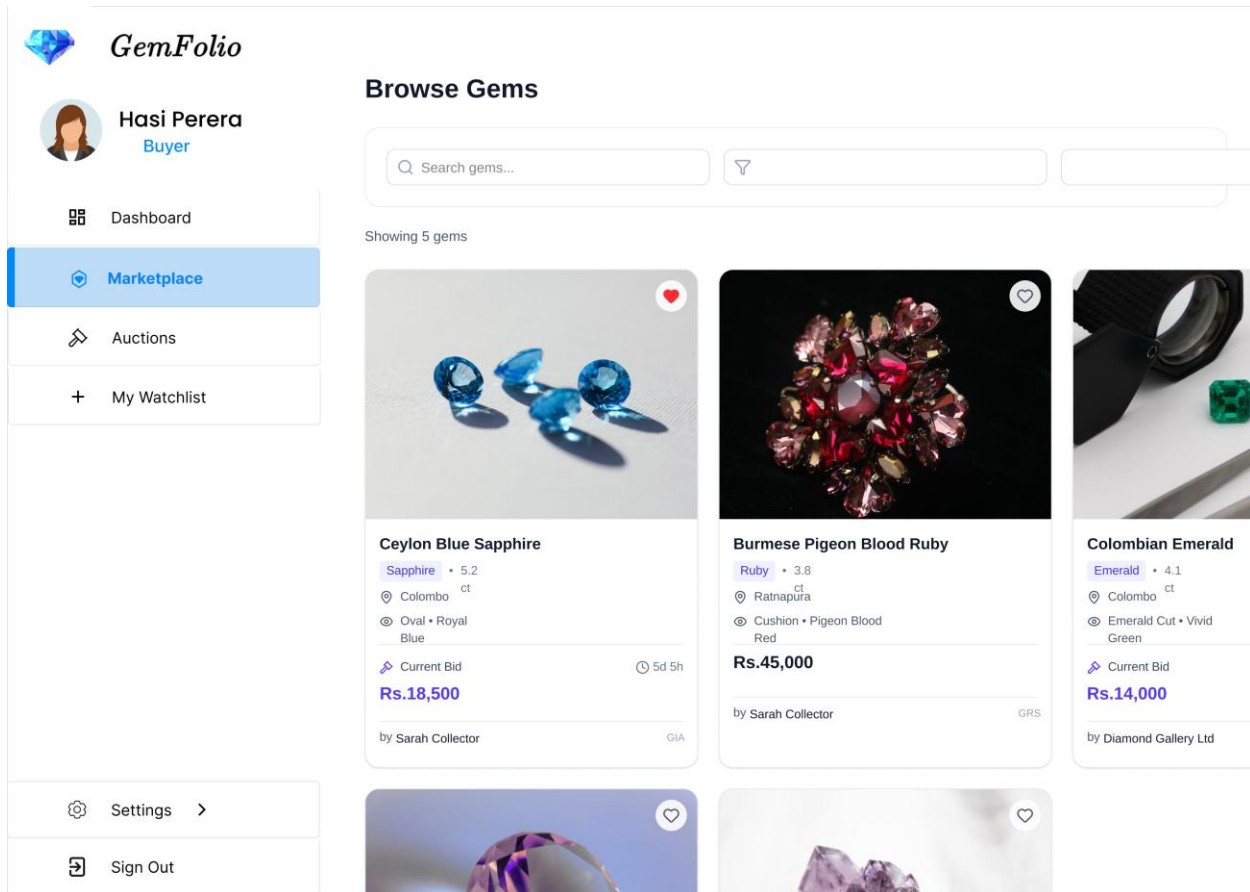


Figure 5:Buyer Marketplace

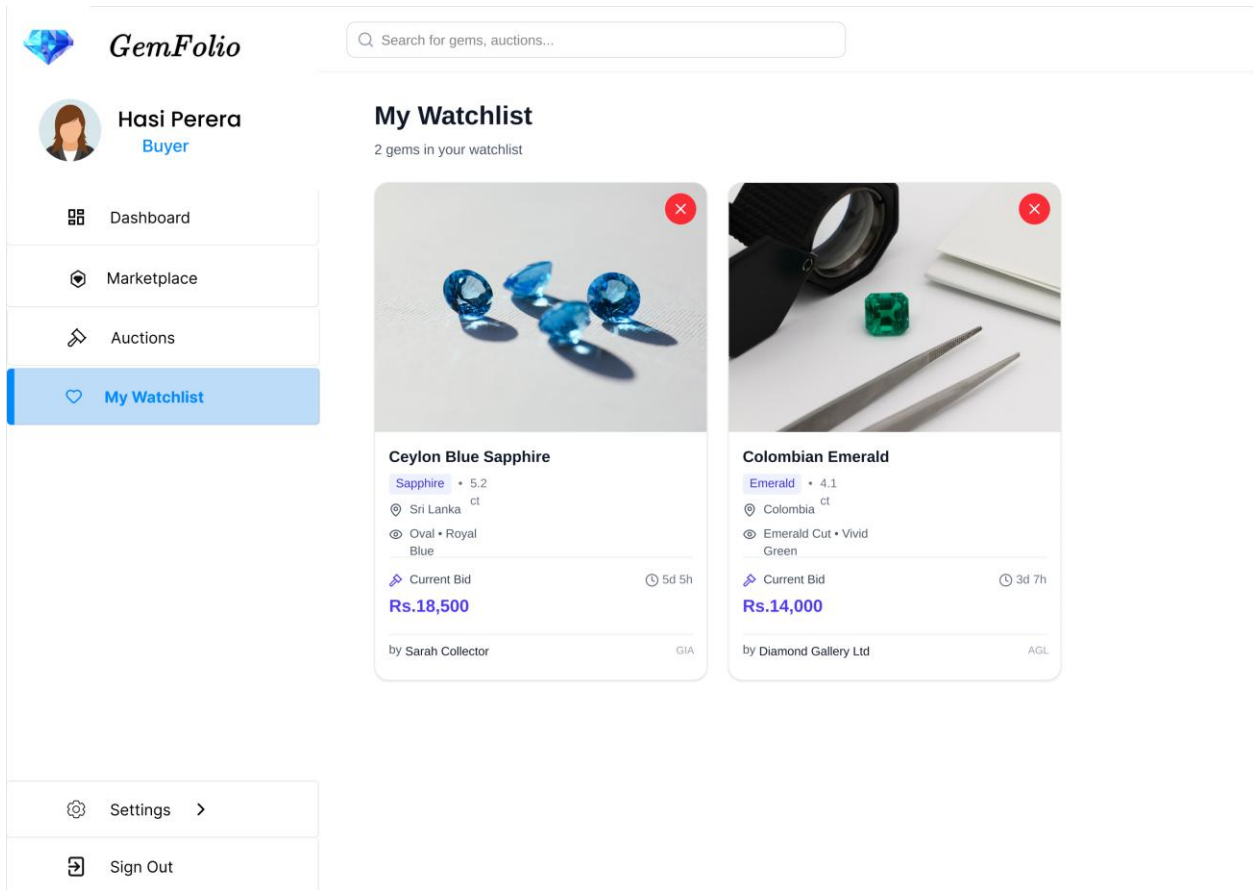


Figure 6:Buyer Watchlist

3.1.3. Administrator Interface:

Administrators have access to a specialized interface for monitoring and managing all platform activities. The admin dashboard provides:

- Verification tools: Admins review gem submissions and attached certificates, ensuring that only authentic listings are approved.
- Approval and rejection controls: Listings can be approved, rejected, or returned to the seller for corrections.
- Auction monitoring: Admins track active auctions, ensuring fair play and resolving disputes if needed.
- User management: Admins can add, modify, or deactivate user accounts and manage roles.
- Reporting and analytics: Admins can generate reports on platform activity, listing performance, and auction trends to support decision-making.

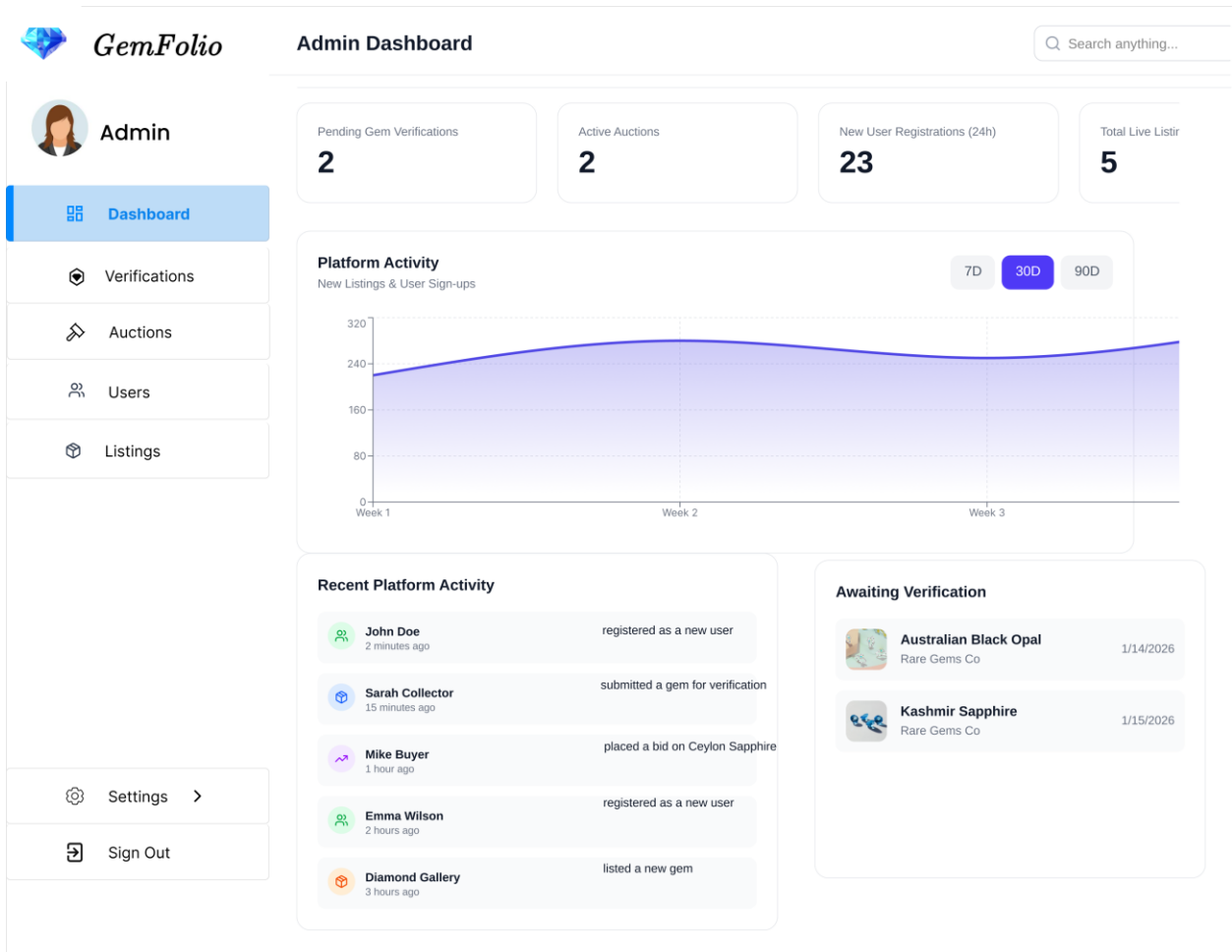


Figure 7: Admin Dashboard


GemFolio


Admin

Dashboard

Verifications

Auctions

Users

Listings

Settings >

Sign Out

User Management

Manage all users on the platform

Total Users
7

Buyers
3

Sellers
3

Admins
1

Figure 8:User Management page

Verify Listing

Review and approve or reject this gem listing



Australian Black Opal

Type	Carat
Opal	8.7 ct
Cut	Color
Freeform Cabochon	Multi-color on Black
Origin	
Lightning Ridge, Australia	
Description	
Magnificent black opal with vibrant play-of-color showing red, blue, and green. Rare lightning ridge specimen with exceptional pattern.	
Listing Type	
Auction	
Starting: Rs.8,000	
Seller	
Rare Gems Co	
ID:	
seller3	

[View Full Resolution](#)

Certificate Document

AGTA Certificate #AG456123

[View Certificate](#)
[Verify with Issuer](#)

Verification Checklist

- ☐ Certificate is authentic and matches gem details
- ☐ Images are clear and show the actual gem
- ☒ Seller profile is complete and verified
- ☐ No signs of tampering or fraud
- ☐ Pricing is reasonable for gem type and quality

[Reject Listing](#)
[Approve Listing](#)

Complete all checklist items to approve this listing

Figure 9:Verify Gem page

3.2 Hardware Interfaces

The system is designed to be platform-independent and compatible with a wide range of hardware devices, including:

- Desktop computers and laptops running modern operating systems such as Windows, macOS, or Linux.
- Tablets and smartphones operating on iOS or Android, supporting responsive web views.

Communication between user devices and the backend server occurs over the internet using standard web protocols. All data transmission is encrypted using HTTPS, ensuring secure and confidential communication. No special hardware peripherals are required; standard input devices such as keyboards, mice, or touchscreens are sufficient for all system operations. The design prioritizes accessibility and ensures consistent experience across devices with varying screen sizes and resolutions.

3.3 Software Interfaces

The system interacts with several software components and external services to provide full functionality:

- Web Browsers: Google Chrome, Mozilla Firefox, and Microsoft Edge are supported for frontend access, ensuring compatibility and optimal performance.
- Database System: MongoDB stores user data, gem details, certificate metadata, auction records, and bidding information in a secure, scalable, and flexible NoSQL environment.
- Cloud Storage Services: External cloud storage platforms, such as AWS S3 or Cloudinary, are used to securely store gem images and certificate files, enabling efficient retrieval and backup.
- Authentication Services: JSON Web Tokens (JWT) handle secure authentication, session management, and role-based authorization.
- Backend Framework: Node.js with Express.js manages API requests, business logic, and communication between frontend and database. This architecture ensures modularity, maintainability, and scalability, allowing future enhancements such as automated certificate validation or payment integration.

Each software interface is selected to ensure high performance, reliability, and ease of maintenance, while supporting future extensibility of the system.

3.4 Communication Interfaces

The Gem Collector's Portfolio and Marketplace System employs standard internet communication protocols to ensure secure, reliable, and efficient data transfer between clients, servers, and external services:

- HTTP/HTTPS: All communication between the user interface and server backend is conducted over HTTPS, ensuring encryption and preventing unauthorized access or data tampering.
- RESTful APIs: The backend exposes RESTful APIs that facilitate frontend interactions, allowing clients to retrieve, submit, update, and delete data as permitted by their role.
- Real-Time Communication (Optional): To support auction functionality, the system may employ WebSockets or similar technologies to enable near real-time bid visibility and notifications. This ensures buyers can place bids and see updates immediately without needing to refresh the page, creating a dynamic and engaging user experience.

The communication interfaces are designed to maintain responsiveness, integrity, and security, ensuring seamless interaction between all system components and external services.

Chapter 4 System Features

This chapter highlights the overall system characteristics of the Gem Collector's Portfolio and Marketplace System. Each system feature presented here lists the system functionality, participants, system behavior, and system flow. It also explains who the participants are and how the system will ensure that there are security and reliability processes implemented throughout the entire gem offering and auctioning system. All system features above aim to ensure that there is a smooth and trusted online experience for the users and that admin-verified processes are in place at all times.

4.1 User Registration and Authentication

4.1.1. Description

The system has a safe user registration and login system for all users. During the registration process, users are required to give basic personal details such as name, email address, and password while choosing their role as a seller/collector or a buyer. Administrators have special accounts that are managed by the system administrator in order to keep system safety.

The authentication mechanism is developed utilizing JSON Web Token technology. This enables secure sessions and safeguards them from various types of attacks like session hijacking and cross-site scripting attacks. After logging in to the system successfully, the user is automatically navigated to a dashboard based on their role. This enables them to view corresponding facilities and notifications based on their role. The proposed system has strict access controls to ensure that unauthorized users are not able to access restricted facilities. This ensures that only authorized users can access facilities like the approval of gems and management of auctions.

Functional Requirements

- FR-1: The system must allow users to register by providing their name, email, password, and role selection.
- FR-2: The system must authenticate users using a secure login mechanism.
- FR-3: The system must enforce role-based access control to restrict access based on user roles.
- FR-4: The system must prevent unauthorized access to restricted features or dashboards.

System Behavior:

- Passwords are encrypted before storage to enhance security.
- Duplicate email registrations are automatically prevented.
- Session tokens expire after periods of inactivity, requiring re-authentication.
- Users are redirected to role-specific dashboards upon login, ensuring they only access relevant functionalities.

4.2 Gem Upload and Portfolio Management

4.2.1. Description

The system has a dedicated dashboard for sellers and collectors to upload the listings of the gems and manage their personal digital portfolio. Each listing of a gem includes specific, detailed structured information such as the type of gem, weight, cut, color, origin, and a summary descriptive overview. High-quality images of the gem and official certification documents are also uploaded for authenticity and to gain buyer confidence.

Every uploaded gem will be kept in the user's portfolio and will be automatically labeled as "Pending Approval" until it gets validated by an administrator. Ensuring only real and certified gems will show up on the public marketplace. Portfolio management capabilities thus enable users to categorize gems, track listing status, and enable or disable the visibility of their particular gems. The proposed enhancement allows sellers and collectors to maintain a professional, organized, and secure digital repository of their certified gems.

4.2.2. Functional Requirements

- FR-5: The system shall allow sellers/collectors to upload gem details including type, weight, origin, and description.
- FR-6: The system shall allow uploading gem images in supported formats.
- FR-7: The system shall allow uploading Gem & Jewelry Authority certificates for verification.
- FR-8: The system shall store gems in a personal portfolio linked to the user account.
- FR-9: The system shall mark newly uploaded gems as “Pending Approval” until verified by administrators.

System Behavior:

- The system validates all required fields before submission to ensure completeness.
- Image and certificate file types are restricted to prevent malware or security risks.
- Users can edit or delete pending listings prior to admin review.
- Portfolio visibility settings allow users to define public or private display, enhancing control over their content.

4.3 Admin Verification and Approval

4.3.1. Description

Administrators have a major role in ensuring the integrity of the marketplace and the trust of clients. The system provides a strong verification system to ensure that all gems available in the marketplace are genuine. Administrators are tasked with checking gems and verifying certificates uploaded.

Rejected adverts are fed back to their sellers with corrective feedback, enabling an amendment of errors or supply of more information. This helps eliminate scams on market platforms, resulting in a safer community for buyers. Administrators record any choices made to track verification activity.

4.3.2. Functional Requirements

- FR-10: The system shall allow admins to view pending gem submissions.
- FR-11: The system shall allow admins to review uploaded certificate documents.
- FR-12: The system shall allow admins to approve or reject gem listings.
- FR-13: The system shall provide feedback on rejected listings.
- FR-14: The system shall display only approved gems in the marketplace.

System Behavior:

- Admin decisions are logged for auditing and transparency.
- Certificates and user-submitted documents are securely stored to prevent tampering.
- Sellers receive email or system notifications regarding approval or rejection status.

4.4 Marketplace Browsing and Search

4.4.1. Description

The marketplace allows customers to search and view only approved listings of gems by admins. The system has advanced filtering tools to assist customers in looking for gems based on type, carat, origin, certification, and seller rating.

Every gem's detail page includes high-quality images, documentation, and information on verified sellers, so that buyers can make well-informed purchasing decisions. In addition, buyers can also add gems they like to a favorite watchlist, which they can refer to later, and track auction activity for gems they are interested in.

4.4.2. Functional Requirements

- FR-15: The system shall display only admin-approved gem listings.
- FR-16: The system shall allow buyers to search for gems by various attributes.
- FR-17: The system shall allow buyers to view gem details and associated certificates.
- FR-18: The system shall display seller profile information, including verification status and portfolio history.

System Behavior:

- Search results updated dynamically as filters are applied.
- Buyers can save gems to a watchlist for easy access.

- Seller profiles include verification badges and historical data for transparency.

4.5 Auction Management

4.5.1. Description

The system will enable auctioning of gems for sale and will allow the creation of time-bound auctions for potential gems for sale. These auctions will have parameters set by the sellers, which will include start price and minimum bid and time duration of the auction.

The system monitors the highest bid, oversees bids, and automatically closes the auction when the end time is reached. Although online payment is not allowed during this stage, the system provides an honest, transparent, and entertaining bidding process.

4.5.2. Functional Requirements

- FR-19: The system shall allow sellers to create auction listings.
- FR-20: The system shall allow sellers to define auction duration and starting price.
- FR-21: The system shall allow buyers to place bids.
- FR-22: The system shall display the highest bid in real time.
- FR-23: The system shall close auctions automatically at the end time.
- FR-24: The system shall declare the highest bidder as the auction winner.

System Behavior:

- Bid amounts are validated to comply with minimum increment rules.
- Auction timers are updated live on all user dashboards.
- Bid history is recorded to maintain transparency and fairness.

4.6 Post-Auction and Communication Support

4.6.1. Description

After an auction has been closed, the system declares the winner and arranges for the buyer and seller to communicate. Because the site doesn't handle payments, contact information is provided in the most secure way possible, so the transaction can be concluded offline.

That means it should cover all the possible auction life cycles, including listing, tracking of bids, and final communication, without losing the trust and privacy of the users.

4.6.2. Functional Requirements

- FR-25: The system shall notify users when an auction ends.
- FR-26: The system shall display winner information.
- FR-27: The system shall share seller contact details with the winning buyer.

System Behavior:

- Auction result notifications are sent via email and dashboard alerts.
- Communication records are logged for audit purposes.

Chapter 5 Other Non-functional Requirements

This chapter will outline the non-functional requirements of Gem Collector's Portfolio and Marketplace system. Unlike functional requirements, non-functional requirements identify how a system should function and the qualities it must meet. Such requirements include those related to system performance, security, safety, usability, scalability, maintainability, and business compliance. Non-functional requirements enable system stakeholders to conceptualize system constraints, functional qualities, and other technical details. Below are more non-functional requirements of Gem Collector's Portfolio and Marketplace system.

5.1 Performance Requirements

The high-performance operation of the Gem Collector's Portfolio and Marketplace System shall be ensured, both under normal conditions and at times of peak usage. The system shall support a very large number of concurrent users on the platform, including sellers, collectors, buyers, and administrators, without major degradation in response time or system stability.

Critical system pages include user registration, login, role-specific dashboards, listings of gems, portfolio views, and auction pages, which should load efficiently. Page load times should not exceed 2–3 seconds under normal operating conditions to keep the interactions responsive and prevent frustration for the end user.

The system should also support real-time updates in regard to ongoing auctions: this means that at any moment, the highest current bid, the bidder, and remaining auction time will need to be updated across all active sessions. Search and filtering operations should return results efficiently, even when the database contains a large number of gem records.

Backend services must be optimized to efficiently handle:

- API requests for retrieving and updating data.
- Image and certificate uploads, including file validation and cloud storage integration.
- Database read/write operations, ensuring transactional integrity and minimal latency.
- Auction bid processing, with immediate reflection of bid updates to all participants.

The system must also be capable of storing and processing high volumes of gem listings, images, and certificate documents without noticeable performance degradation, ensuring scalability and long-term reliability.

5.2 Safety Requirements

The system also needs to provide for the safety and integrity of user data, uploaded files, as well as system operations. The system protects sensitive data such as personal information, login, and contact details from unauthorized access through accidental viewing. The system adheres to the best practices of secure data handling.

Regular automated database backups must be performed to prevent data loss due to:

- Server failures or crashes.
- Data corruption or system errors.
- Accidental deletion by users or administrators.
- Cybersecurity incidents such as ransomware or hacking attempts.

The uploaded gem certificates and images will be stored in secure cloud storage with restricted access permissions. Access to the certificate files for verification purposes is allowed only to authorized administrators.

The system imposes a strict verification process for gem listings, in which only administrators are authorized to accept or reject gem listings. Thus, unverified, tampered with, or fraudulent gems will not reach the buyers at all.

5.3 Security Requirements

The system implements robust security measures to protect user accounts, system resources, and sensitive data. Key security features include:

- **Strong Authentication:** Users must log in using a valid email or username combined with a secure password. Passwords are hashed and salted before storage, preventing unauthorized access in case of data breaches.
- **Encrypted Communication:** All frontend-backend communication is encrypted using HTTPS, ensuring data confidentiality and integrity during transmission.
- **Session Management and Role-Based Access Control:** JSON Web Tokens (JWT) are used to manage sessions, enable token-based authentication, and enforce role-based authorization.

Authorization Rules:

- Sellers and collectors can only access and manage their own gem listings and personal portfolios.
- Buyers can only browse admin-approved listings and participate in auctions.
- Administrators have exclusive access to verification, moderation, and system management features, ensuring marketplace security.

Sensitive user data, certificate files, and system audit logs are restricted to authorized administrators, preventing unauthorized access or tampering.

5.4 Software Quality Attributes

The system is designed to meet high-quality software standards, ensuring that users of varying technical expertise can use the platform efficiently. Key software quality attributes include:

- **Usability:** The interface is intuitive, consistent, and easy to navigate, reducing the learning curve for sellers, buyers, and administrators. Tooltips, validation messages, and help documentation enhance the user experience.
- **Maintainability:** The modular architecture of the MERN stack enables developers to apply bug fixes, security patches, and feature enhancements efficiently without disrupting other system components.
- **Scalability:** The platform supports increasing numbers of users, gem listings, portfolios, and auctions without requiring major architectural changes. New features can be integrated seamlessly.
- **Reliability and Availability:** The system ensures stable operation with minimal downtime, high system availability, and redundancy measures to maintain service continuity.

These attributes collectively ensure that the system is user-friendly, robust, and capable of supporting long-term growth in the online gem trading ecosystem.

5.5 Business Rules

To maintain marketplace integrity, trust, and regulatory compliance, the system follows several mandatory business rules:

- Only administrators have the authority to approve or reject gem listings.
- Only admin-approved gems are displayed in the marketplace and made available for auctions.
- Sellers and collectors must upload valid certificates issued by recognized authorities for each gem listing.
- Buyers can only view and bid on verified listings, ensuring fairness and authenticity.
- Auction listings must define a start price, minimum bid increment, and duration to maintain transparent and structured bidding.
- Online payment processing is excluded; transactions must be completed externally between buyers and sellers.
- Sellers are responsible for providing accurate and up-to-date gem information and certificates.
- Any attempt to upload uncertified, fraudulent, or manipulated gems may result in account suspension or permanent restriction.

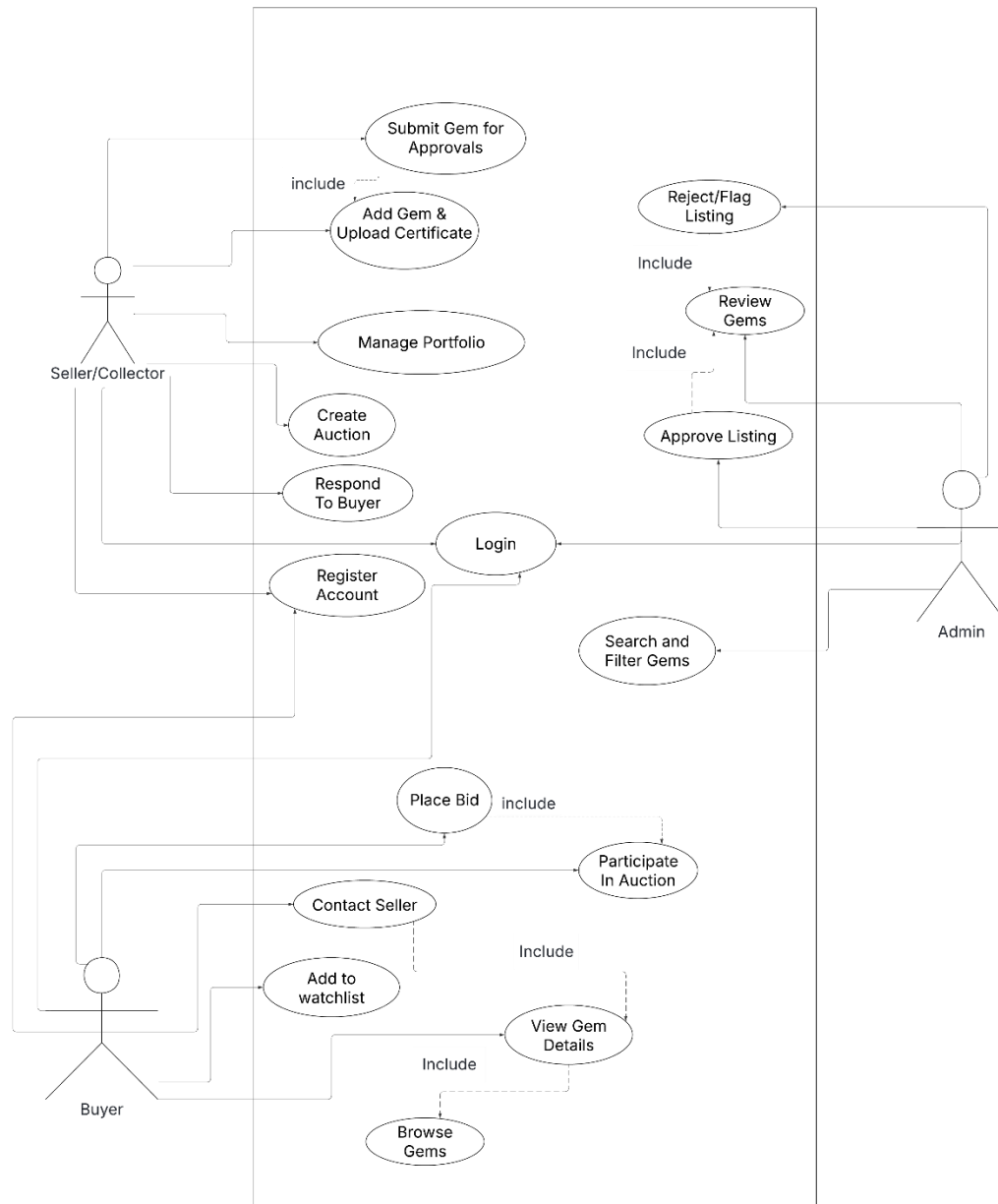
These rules enforce ethical trading practices and provide clear boundaries for user interactions, ensuring a trustworthy marketplace environment for all participants.

Chapter 6 Appendix A – Glossary

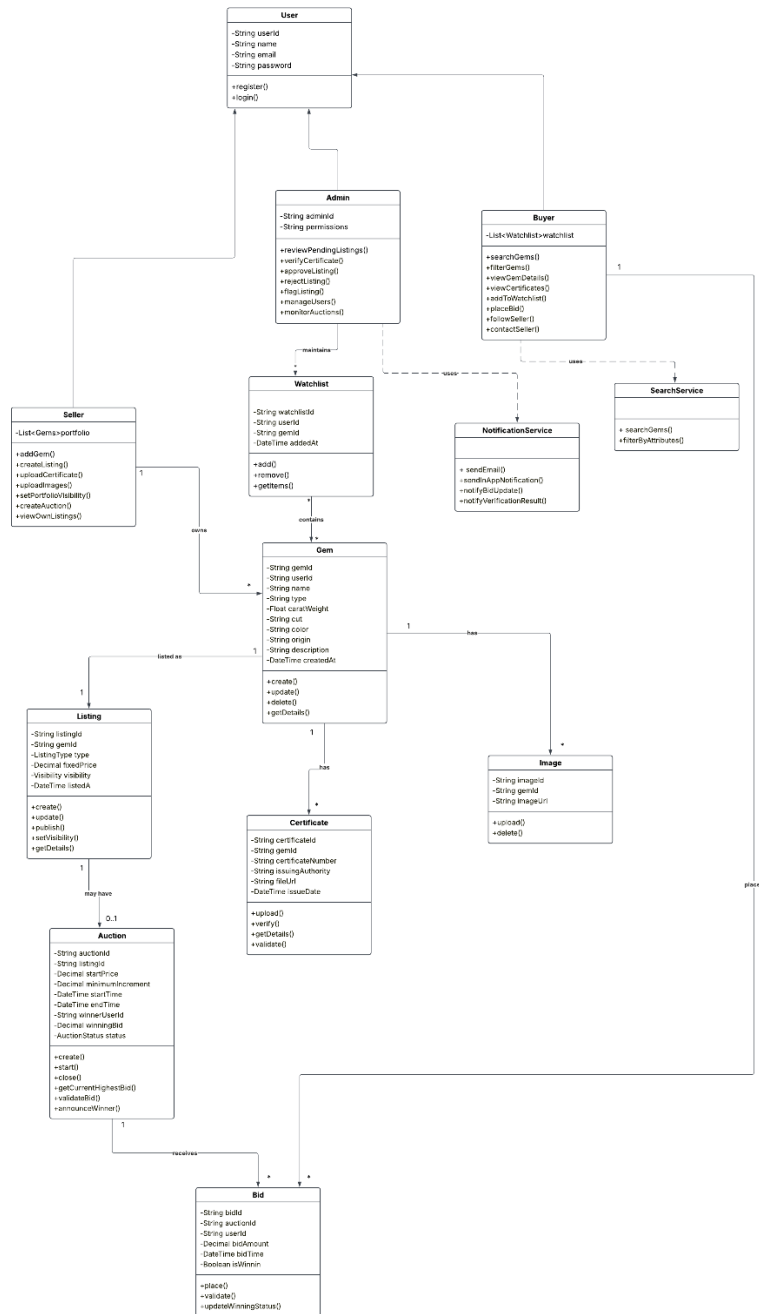
Term	Description
Admin	A system user responsible for verifying gem listings, approving certificates, monitoring auctions, and managing platform operations.
Auction	A time-based bidding process where buyers place bids on a gem and the highest bidder wins when the auction ends.
Authentication	The process of verifying a user's identity before granting access to the system.
Authorization	The process of granting role-based access permissions to users.
Buyer	A user who browses verified gem listings and participates in auctions.
Certificate	An official document issued by a recognized authority verifying a gem's authenticity and characteristics.
Cloud Storage	An online service used to store gem images and certificate files securely.
Collector	A user who maintains a personal portfolio of certified gems on the platform.
Dashboard	A user interface that displays personalized tools and information based on the user's role.
Gem Listing	A record containing gem details, images, certificates and sale or auction status.
JWT (JSON Web Token)	A secure token-based authentication method used for session management and role-based access control.
Marketplace	The public section of the system where approved gem listings are displayed to buyers.
MERN Stack	A web development stack consisting of MongoDB, Express.js, React.js and Node.js.
Portfolio	A personal digital collection of gems managed by a seller or collector.
Role-Based Access Control	A security mechanism that restricts system access based on the user roles.
Seller	A user who uploads certified gems for sale or auction.
Verification	The process is performed by administrators to validate gem details and certificates before public display.

Chapter 7 Appendix B – Analysis Models

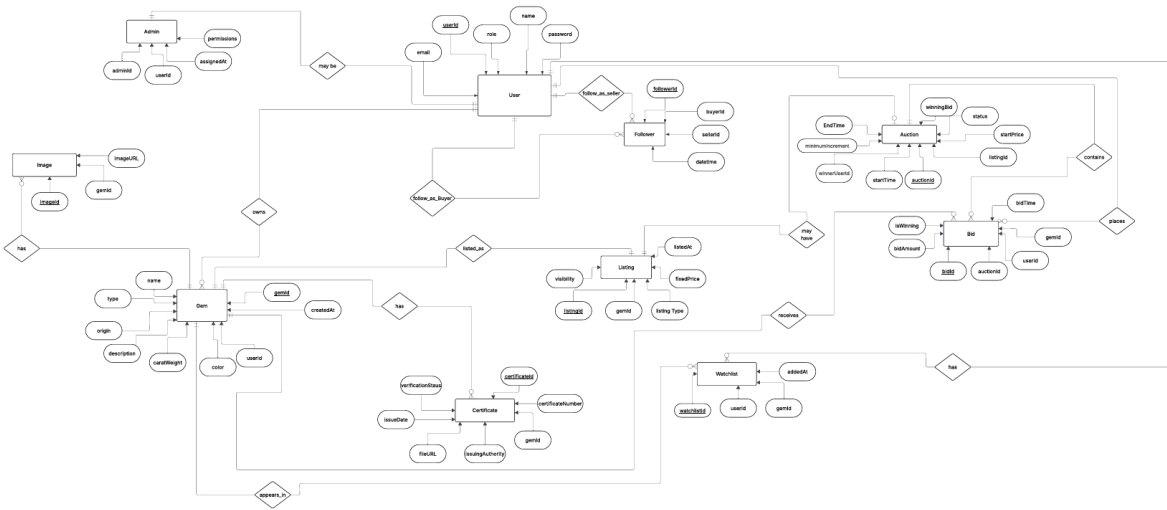
Use Case Diagram



Class Diagram



ER Diagram



Note: Due to page size limitations, these diagrams are provided as reference images.

The full-resolution diagrams are available in Google Drive:

<https://drive.google.com/drive/folders/1-TBiKIOvcvmLUFk8so1Xh0fzJBTZ-OLo?usp=sharing>